Lab 01 – Worksheet

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Note: Assumptions and logics should be explained separately in tasks after the task results.

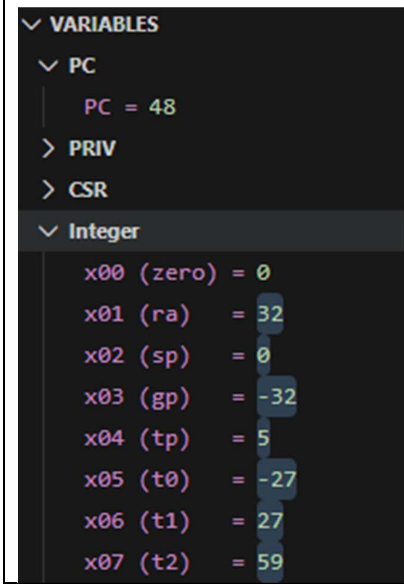
Task 3.

Provide appropriately commented codes (Mention question part before each part)

```
li x1,5 #a=5
addi x2,x0,0 #b=0+0
addi x1,x2,32 #a=b+32
add x4,x1,x2 #temp1=a+b
addi x6,x4,-5 #d=temp1 - 5
sub x4,x1,x6 #temp1=a-d
sub x3,x2,x1 # temp2=b-a
add x5,x4,x3 #temp3=temp1+temp2
add x7,x5,x6 #e=temp3+d
add x7,x7,x1
add x7,x7,x2
add x7,x7,x6

end:
    j end
```

Add screenshot of your results (register)



Task 4a

Write the RISC-V assembly code for each item below. Try guessing the result in each destination before executing the instruction and corroborate it after execution:

- Store x10 as unsigned integer at address 0x100.

```
li x10,0x78786464
sw x10, 0x100(x0)

end:
    j end
```

- Store x11 as unsigned integer at address 0x1F0.

```
li x11,0xA8A81919
sw x11, 0x1F0(x0)

end:
    j end
```

- Load an unsigned short integer (two bytes) from address 0x100 in x12.

```
lhu x12, 0x100(x0)
```

- Load a short integer from address 0x1F0 in register x13.

```
lh x13, 0x1F0(x0)
```

- Load a singed character from address 0x1F0 in register x14.

```
lb x14, 0x1F0(x0)
```

Add screenshots of your results (register & memory both)

Registers:

```
∨ Integer
    x00 (zero) = 0x00000000
    x01 (ra)   = 0x00000000
    x02 (sp)   = 0x7FFFFFF0
    x03 (gp)   = 0x10000000
    x04 (tp)   = 0x00000000
    x05 (t0)   = 0x00000000
    x06 (t1)   = 0x00000000
    x07 (t2)   = 0x00000000
    x08 (s0)   = 0x00000000
    x09 (s1)   = 0x00000000
    x10 (a0)   = 0x78786464
    x11 (a1)   = 0xA8A81919
    x12 (a2)   = 0x00006464
    x13 (a3)   = 0x00001919
    x14 (a4)   = 0x00000019
```

Memory:

0x00000100	64	64	78	78
0x000001F0	19	19	A8	A8

Task 4b

Provide appropriately commented codes (Mention question part before each part)

```
    #Base addresses
    li x10, 0x100 #a(char array)
    li x11, 0x200 #b(short array)
    li x12, 0x300 #c(unsigned integer array)

    #adding dummy values to each array to show wroking
    # char array a = [1,2,3,4]
    li x5,1
    sb x5,0(x10)
    li x5,2
    sb x5,1(x10)
    li x5,3
    sb x5,2(x10)
    li x5,4
    sb x5,3(x10)
```

```
    # short array b = [10,20,30,40]
    li x5,10
    sh x5,0(x11)
    li x5,20
    sh x5,2(x11)
    li x5,30
    sh x5,4(x11)
    li x5,40
    sh x5,6(x11)
    # unsigned int array c = [0,0,0,0]
    sw x0,0(x12)
    sw x0,4(x12)
    sw x0,8(x12)
    sw x0,12(x12)

    #i=0
    lb x13, 0(x10) #a[0]
    lh x14, 0(x11) #b[0]
    add x15, x13, x14 #a[0]+b[0]
    sw x15, 0(x12) #c[0]

    #i=1
    lb x13, 1(x10)    # a[1]
    lh x14, 2(x11)    # b[1]
    add x15, x13, x14
    sw x15, 4(x12)    # c[1]


    #i=2
    lb x13, 2(x10) #a[2]
    lh x14, 4(x11) #b[2]
    add x15, x13, x14
    sw x15, 8(x12) #c[2]


    #i=3
    lb  x13, 3(x10) #a[3]
    lh  x14, 6(x11) #b[3]
    add x15, x13, x14
    sw  x15, 12(x12) #c[3]

end:
    j end
```

Add screenshots of your results (register & memory both)

Char array a:

0x00000100	1	2	3	4

Short array b:

0x00000204	30	0	40	0
0x00000200	10	0	20	0

Unsigned int array:

0x0000030C	44	0	0	0
0x00000308	33	0	0	0
0x00000304	22	0	0	0
0x00000300	11	0	0	0

Registers:

```
∨ VARIABLES
  ∨ PC
      PC = 0x0000009C
  > PRIV
  > CSR
  ∨ Integer
      x00 (zero) = 0x00000000
      x01 (ra)   = 0x00000000
      x02 (sp)   = 0x7FFFFFF0
      x03 (gp)   = 0x10000000
      x04 (tp)   = 0x00000000
      x05 (t0)   = 0x00000028
      x06 (t1)   = 0x00000000
      x07 (t2)   = 0x00000000
      x08 (s0)   = 0x00000000
      x09 (s1)   = 0x00000000
      x10 (a0)   = 0x00000100
      x11 (a1)   = 0x00000200
      x12 (a2)   = 0x00000300
      x13 (a3)   = 0x00000004
      x14 (a4)   = 0x00000028
      x15 (a5)   = 0x0000002C
```

Lab 01
Assessment Rubrics

Task No.	LR 2 Code	LR 5 Results	AR 7 Report Submission/Code Comments
Task 1a	/05	/05	/20
Task 1b	/10	/05	
Task 2a	/10	/05	
Task 2b	/20	/05	
Exercise	/10	/05	
Total Points	/100 Points		
CLO Mapped	CLO 2		

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	**Program/Code/ Simulation Model/ Network Model**	Program/code/simulation model/network model doesnot implement the requiredfunctionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulationmodel/network model has some errors and doesnot produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulationmodel/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full commandon the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorlyconstructed with erroneousresults. Titles, captions, units are not mentioned. Data is presented in anobscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are notaccurate. Data presentation is not tooclear.	All figures, graphs, tables are correctly drawn but contain minorerrors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captionsand proper units are mentioned. Data presentation is systematic.
AR9	**Report Content/Code Comments**	No summary provided. The number/amount of tasks completed below the level of satisfaction and/or submitted late	Couldn't provide good summary of in-lab tasks. Some major tasks were completed but not explained well. Submission on time. Some major plots andfigures provided	Good summary of In-lab tasks. All major tasks completed except few minor ones. The work is supported by some decent explanations, Submission on time, All necessary plots, and figures provided	Outstanding Summary of In-Lab tasks. All task completed and explained well, submitted on time, good presentation of plotsand figure with proper label, titles and captions